KAI CHEN

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RESEARCH OVERVIEW

My research aims at constructing reliable **Multi-modal** AI systems from a **data-centric** perspective. Recently, we have witnessed the interim success of training foundational models on massive **human data**, which, however, is believed to come to an end. Towards the second half of AI, a scalable **synthetic data** generation and training pipeline is necessary, for which **reinforcement learning** serves as a well-formulated solution. Thus, **Scaling Reinforcement Learning for (M)LLMs** has become indispensable for achieving Artificial Super-Intelligence (ASI), which has obtained remarkable success for text-only LLMs, while remaining open challenges for (native) **Multi-modal LLMs**. Currently, I'm trying to answer the following questions,

- **Policy:** *How to build end-to-end Multi-modal LLMs with frontier visual, textual, and speech abilities?*
- **World:** *How to construct 3D visual world models in a controllable and scalable manner?*
- **Feedback:** *How to get effective intrinsic feedback from (M)LLMs themselves without reward models?*
- **Optimization:** *Does more data always result in better performance?*

Research Areas: Omni-modal LLMs, Visual World Modeling, Mixture-of-Experts (MoE)

EDUCATION

Hong Kong University of Science and Technology, Hong Kong SAR — *Sep 2020 - Oct 2026*
Ph.D. in **Computer Science and Engineering**
GPA: 4.10/4.0
Advisor: Prof. Dit-Yan Yeung

Fudan University(FDU), Shanghai, China — *Sep 2016 - June 2020*
B.S. in **Computer Science**, Minor in **Economics** (Outstanding Graduates of Shanghai)
Overall GPA: 3.70/4.0, Major GPA: 3.90/4.0, Ranking: 3/32
Advisor: Prof. Yanwei Fu

University of Manchester, Manchester, UK — *Sep 2018 - Jan 2019*
Exchange student in the **Department of Computer Science**
Advisor: Dr. Tingting Mu

EXPERIENCE

Mobile Intelligence Group (MIG), SenseTime — Oct 2019 - April 2020
Research Intern — *Advisor:Dr. Wenxiu Sun*

- Research on real-time (portrait) instance segmentation deployable on mobile devices.

Computer Vision Lab, Indiana University Bloomington (IUB) — June 2019 - Sep 2019
Global Talent Attraction Program (GTAP) Visiting Scholar — *Advisor:Prof. David Crandall*

- Research on semi-supervised semantic segmentation and indoor 3D reconstruction.

SELECTED HONORS

ICML 2026 Silver Reviewer Awards	May 2026
CVPR 2025 Doctoral Consortium Awards	May 2025
HKUST Research Travel Grant	2023-2025
HKUST Postgraduate Scholarship	Sep 2020
Outstanding Graduates of Shanghai [Wechat Post] (5%, by Shanghai Government)	April 2020

Scholarship for Outstanding Graduates (5%, by Fudan University) April 2020
Oversea Visiting Student Stipend of (15,000 CNY, Fudan University) Dec 2019
Joel & Ruth Spira Scholarship (1%, by Lutron Electronics) Mar 2019
National Scholarship (1%, by Ministry of Education of P.R.China) Sep 2018
Scholarship for Outstanding Undergraduate Students (5%, by Fudan University) Oct 2017

PUBLICATIONS

Full publication list on my Google Scholar. (* denotes equal contribution)

I. Multi-modal Foundation Models - Omni-modality and Reasoning

RQ: How to construct multi-modal LLMs with visual, textual, and speech reasoning abilities simultaneously?

[C24] **EMOVA: Empowering Language Models to See, Hear and Speak with Vivid Emotions** **CVPR 2025**
Kai Chen*, Yunhao Gou*, Runhui Huang*, Zhili Liu*, Daxin Tan*, *and other 26 authors* [link]

[C23] **Reasoning-Aligned Perception Decoupling for Scalable Multi-modal Reasoning** **ICLR 2026**
Yunhao Gou*, **Kai Chen***, Zhili Liu*, Lanqing Hong, Xin Jin, Zhenguo Li, [link]
James T. Kwok, Yu Zhang.

[C22] **Speech-Omni-Lite: Portable Speech Interfaces for Vision-Language Models** **Arxiv 2026**
Dehua Tao, Xuan Luo, Daxin Tan, **Kai Chen**, Lanqing Hong, Jing Li, Ruifeng Xu, Xiao Chen [link]

[C21] **AEQ-Bench: Measuring Empathy of Omni-Modal Large Models** **ACL 2026 Findings**
Xuan Luo, Lewei Yao, Libo Zhao, Lanqing Hong, **Kai Chen**, Dehua Tao, Daxin Tan [link]
Ruifeng Xu, Jing Li.

II. Multi-modal Foundation Models - Mixture of Cluster-conditional Experts (MoCE)

RQ: Does more data always result in better performance during model pre-training and fine-tuning?

[J3] **Mixture of Cluster-conditional LoRA Experts for Vision-language Instruction Tuning** **IEEE TIP 2026**
Yunhao Gou*, Zhili Liu*, **Kai Chen***, Lanqing Hong, Hang Xu, Aoxue Li, [link]
Dit-Yan Yeung, James Kwok, Yu Zhang.

[C20] **Task-customized Masked Autoencoder via Mixture of Cluster-conditional Experts** **ICLR 2023 Spotlight**
Zhili Liu*, **Kai Chen***, Jianhua Han, Lanqing Hong, Hang Xu, Zhenguo Li, James Kwok. [link]

[C19] **Task-Customized Self-Supervised Pre-training with Scalable Dynamic Routing** **AAAI 2022**
Zhili Liu, Jianhua Han, **Kai Chen**, Lanqing Hong, Hang Xu, Chunjing Xu, Zhenguo Li. [link]

III. Multi-modal Foundation Models - Scalable Oversight for (M)LLM Self-alignment

RQ: Are there any intrinsic scalable oversight from (M)LLMs to supervise their own capabilities?

[C18] **Corrupted but Not Broken: Rethinking the Impact of Corrupted Data in Visual Instruction Tuning** **EMNLP 2025 Oral**
Yunhao Gou, Hansi Yang, Zhili Liu, **Kai Chen**, Yihan Zeng, Lanqing Hong, Zhenguo Li, [link]
Qun Liu, James T Kwok, Yu Zhang.

[J2] **Unified Triplet-Level Hallucination Evaluation for Large Vision-Language Models** **TMLR 2025**
Junjie Wu*, Tsz Ting Chung*, **Kai Chen***, Dit-Yan Yeung. [link]

[C17] **Mixture of insighTful Experts (MoTE): The Synergy of Thought Chains and Expert Mixtures in Self-Alignment** **ACL 2025**
Zhili Liu*, Yunhao Gou*, **Kai Chen***, Lanqing Hong, Jiahui Gao, Fei Mi, Yu Zhang, [link]
Zhenguo Li, Xin Jiang, Qun Liu, James T. Kwok.

[C16] **Eyes Closed, Safety On: Protecting Multimodal LLMs via Image-to-Text Transformation** **ECCV 2024**
Yunhao Gou*, **Kai Chen***, Zhili Liu*, Lanqing Hong, Hang Xu, Zhenguo Li, Dit-Yan Yeung, James Kwok, Yu Zhang. [link]

[C15] **Gaining Wisdom from Setbacks: Aligning Large Language Models via Mistake Analysis** **ICLR 2024**
Kai Chen*, Chunwei Wang*, Kuo Yang, Jianhua Han, Lanqing Hong, Fei Mi, Hang Xu, Zhengying Liu, Wenyong Huang, Zhenguo Li, Dit-Yan Yeung, Lifeng Shang, Xin Jiang, Qun Liu. [link]

IV. Visual World Models - Corner Cases for Autonomous Driving (CODA)

RQ: How to enhance the robustness of self-driving agents towards road corner cases?

A: 1) corner case collection, 2) corner case generation, and 3) multi-modal reasoning

[C14] **ECCV 2024 W-CODA: 1st Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving** **ECCV 2024**
Kai Chen*, Ruiyuan Gao*, Lanqing Hong*, Hang Xu, Xu Jia, Holger Caesar, Dengxin Dai, Bingbing Liu, Dzmitry Tsishkou, Songcen Xu, Chunjing Xu, Qiang Xu, Huchuan Lu, Dit-Yan Yeung. [link]

[C13] **CODA-LM: Automated Evaluation of Large Vision-Language Models on Self-driving Corner Cases** **WACV 2025**
Kai Chen*, Yanze Li*, Wenhua Zhang*, Yanxin Liu, Pengxiang Li, Ruiyuan Gao, Lanqing Hong, Meng Tian, Xinhai Zhao, Zhenguo Li, Dit-Yan Yeung, Huchuan Lu, Xu Jia. [link]

[C12] **CODA: A Real-World Road Corner Case Dataset for Object Detection in Autonomous Driving** **ECCV 2022**
Kaican Li*, **Kai Chen***, Haoyu Wang*, Lanqing Hong, Chaoqiang Ye, Jianhua Han, Yukuai Chen, Wei Zhang, Chunjing Xu, Dit-Yan Yeung, Xiaodan Liang, Zhenguo Li, Hang Xu. [link]

V. Visual World Models - Geometric-controllable Visual Generation

RQ: How to generate the 3D visual world in a controllable and scalable manner?

[C11] **MagicDrive3D: Controllable 3D Generation for Any-View Rendering in Street Scenes** **WACV 2026**
Ruiyuan Gao, **Kai Chen**, Zhihao Li, Lanqing Hong, Zhenguo Li, Qiang Xu. [link]

[C10] **MagicDrive-V2: High-Resolution Long Video Generation for Autonomous Driving with Adaptive Control** **ICCV 2025**
Ruiyuan Gao, **Kai Chen**, Bo Xiao, Lanqing Hong, Zhenguo Li, Qiang Xu. [link]

[C9] **Implicit Concept Removal of Diffusion Models** **ECCV 2024**
Zhili Liu*, **Kai Chen***, Yifan Zhang, Jianhua Han, Lanqing Hong, Hang Xu, Zhenguo Li, Dit-Yan Yeung, James Kwok. [link]

[C8] **DetDiffusion: Synergizing Generative and Perceptive Models for Enhanced Data Generation and Perception** **CVPR 2024**
Yibo Wang*, Ruiyuan Gao*, **Kai Chen***, Kaiqiang Zhou, Yingjie Cai, Lanqing Hong, Zhenguo Li, Lihui Jiang, Dit-Yan Yeung, Qiang Xu, Kai Zhang. [link]

[C7] **MagicDrive: Street View Generation with Diverse 3D Geometry Control** **ICLR 2024**
Ruiyuan Gao*, **Kai Chen***, Enze Xie, Lanqing Hong, Zhenguo Li, Dit-Yan Yeung, Qiang Xu. [link]

[C6] **TrackDiffusion: Tracklet-Conditioned Video Generation via Diffusion Models** **WACV 2025**
Pengxiang Li*, **Kai Chen***, Zhili Liu*, Ruiyuan Gao, Lanqing Hong, Dit-Yan Yeung, Huchuan Lu, Xu Jia. [link]

[C5] **GeoDiffusion: Text-Prompted Geometric Control for Object Detection Data Generation** **ICLR 2024**
Kai Chen*, Enze Xie*, Zhe Chen, Yibo Wang, Lanqing Hong, Zhenguo Li, Dit-Yan Yeung. [link]

VI. Representation Learning - Object-level Self-supervised Learning (SSL)

RQ: How to perform object-level SSL for better transferability on downstream dense perception tasks?

[C4] **Mixed Autoencoder for Self-supervised Visual Representation Learning** **CVPR 2023**
Kai Chen*, Zhili Liu*, Lanqing Hong, Hang Xu, Zhenguo Li, Dit-Yan Yeung. [link]

[C3] **MultiSiam: Self-supervised Multi-instance Siamese Representation Learning for Autonomous Driving** **ICCV 2021**
Kai Chen, Lanqing Hong, Hang Xu, Zhenguo Li, Dit-Yan Yeung. [link]

[C2] **SODA10M: A Large-Scale 2D Self/Semi-Supervised Object Detection Dataset for Autonomous Driving** **NeurIPS 2021**
Jianhua Han, Xiwen Liang, Hang Xu, **Kai Chen**, Lanqing Hong, Jiageng Mao, Chaoqiang Ye, Wei Zhang, Zhenguo Li, Xiaodan Liang, Chunjing Xu. [link]

Early Works

[J1] **Automatic Dense Annotation for Monocular 3D Scene Understanding** **IEEE Access 2020**
Md. Alimoor Reza, **Kai Chen**, Akshay Naik, David Crandall, Soon-Heung Jung. [link]

[C1] **Automatic Annotation for Semantic Segmentation in Indoor Scenes** **IROS 2019**
Md Alimoor Reza, Akshay Naik, **Kai Chen**, David Crandall. [link]

ACADEMIC SERVICES

Program Committee / Organizer

- The 1st W-CODA Workshop at ECCV 2024 on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving. 2024
- The 2nd SSLAD workshop at ECCV 2022. 2022
- The 1st SSLAD workshop at ICCV 2021 on Self-supervised Learning for Next-generation Industry-level Autonomous Driving. 2021

Area Chair

- International Joint Conferences on Artificial Intelligence (IJCAI) 2025

Conference Reviewer

- IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2022-2026
- IEEE International Conference on Computer Vision (ICCV) 2023-2025
- European Conference on Computer Vision (ECCV) 2022-2024
- ACL Rolling Review (ARR) 2025-2026
- International Conference on Learning Representations (ICLR) 2023-2026
- International Conference on Machine Learning (ICML) 2025-2026
- Neural Information Processing Systems (NeurIPS) 2021-2025
- International Joint Conferences on Artificial Intelligence (IJCAI) 2023-2025
- AAAI Conference on Artificial Intelligence (AAAI) 2022
- International Conference on Robotics and Automation (ICRA) 2022
- ACM International Conference on Multimedia (ACM MM) 2025
- IEEE Winter Conference on Applications of Computer Vision (WACV) 2026

Journal Reviewer

- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- IEEE Transactions on Image Processing (TIP)
- IEEE Transactions on Multimedia (TMM)
- IEEE Access

PATENTS

- [CN116665219A] **GeoDiffusion: Text-Prompted Geometric Control for Object Detection Data Generation**. Enze Xie, **Kai Chen**, Lanqing Hong, Zhenguo Li. *Published in May 26th, 2023.*
- [CN115731530A] **MultiSiam: Self-supervised Multi-instance Siamese Representation Learning for Autonomous Driving**. **Kai Chen**, Lanqing Hong, Hang Xu, Zhenguo Li. *Published in Aug. 24th, 2021.*

TEACHING

- **HKUST** COMP 4211 - Machine Learning, Teaching Assistant, Fall 2025.
- **HKUST** COMP 2012 - Object-Oriented Programming and Data Structures, Teaching Assistant, Fall 2024.
- **HKUST** COMP 2012 - Object-Oriented Programming and Data Structures, Teaching Assistant, Fall 2021.
- **HKUST** COMP 2012 - Object-Oriented Programming and Data Structures, Teaching Assistant, Spring 2021.

INVITED TALKS

- [AI TIME Online] EMOVA: Empowering Language Models to See, Hear and Speak with Vivid Emotions. [Recording]
- [VALSE Webinar] Geometric-controllable Visual Generation: A Systemetic Solution.[Recording]
- [AIDriver Online] Controllable Corner Case Generation for Autonomous Driving.[Recording]
- [AI TIME Online] Gaining Wisdom from Setbacks: Aligning Large Language Models via Mistake Analysis. [Recording]
- [TechBeat Online] Gaining Wisdom from Setbacks: Aligning Large Language Models via Mistake Analysis. [Recording]
- [VALSE 2023@Wuxi] Mixed Autoencoder for Self-supervised Visual Representation Learning. [Recording]
- [VALSE 2023@Wuxi] CODA: A Real-World Road Corner Case Dataset for Object Detection in Autonomous Driving. [Recording]

TECHNICAL SKILLS

Program Languages	Python, Matlab, C/C++/C#, SQL, LaTeX
Framework	Pytorch, Tensorflow
Language	Native in Mandarin, Fluent in English and Japanese CET-4(649), CET-6(619), TOEFL-iBT(101)